Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software engineer experienced in **C, C++, and Python**, with systems coursework plus cross-platform integration, automated testing and validation, and performance optimization of software subsystems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Architected a multi-tenant platform on Azure SQL and PostgreSQL for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through query optimization and performance tuning.
- Implemented Python FastAPI data pipelines processing 5,000+ monthly jobs with validation, reconciliation, and CI/CD-backed releases, achieving 99%+ reliability while isolating failures across integrated production subsystems at scale.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Supported analysis, design, build, testing, and deployment of React and Flask product workflows, serving 150+ instructors and processing 10,000+ daily requests with stable production performance at scale.
- Identified and resolved OCR, model, and API deficiencies, implementing automated tests and documented fallback paths while reducing customer feedback turnaround time 60% during integration and validation at scale.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered Java and Python microservices with automated VM testing and 20+ unit and integration test suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS at scale.
- Optimized large-scale API integrations using Python, AWS Glue, and S3, improving response times 30% for low-bandwidth users while reducing technical debt 20% across production ETL workflows end to end.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed a React and Node.js platform integrating eight microservices for authentication, allocation, billing, and notifications, syncing two million records while reducing resource wastage 42% end to end.
- Automated Jenkins testing and Docker-based deployments, cutting deployment time 50% while using Sentry monitoring to identify and resolve production deficiencies during integration and validation in production environments.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Analyzed client requirements and system specifications, mapped 12+ integration points, and presented architecture recommendations in design reviews before implementing cross-platform booking and payment experiences for end users at scale.
- Delivered Python Flask and Firebase solutions across 50,000+ records, validating implementation against specifications while achieving sub-200-millisecond responses and reducing manual refund handling 70% for customers at production scale.

Skills

Languages: C, C++, Python, Java, JavaScript, SQL
Systems: Linux, Git, Bash, Docker, Kubernetes
Testing: Jenkins, PyTest, Mockito, CI/CD, Integration Testing
Backend: FastAPI, Flask, REST APIs, Node.js
Data: PostgreSQL, MySQL, MongoDB, AWS, Azure
Practices: Performance Optimization, System Design, Agile, Debugging, GitHub Actions

Projects

CPU-scheduling Simulator Live | *C++, Operating Systems*
- Implemented C++ scheduling algorithms including FCFS, SJF, SRT, and Round Robin with React controls and Gantt visualization, reducing average turnaround time 25% across 1,000+ simulated process combinations.

Chat2SQL Code | *Cross-Platform Integration*
- Designed Java and TypeScript services supporting 10+ databases, configuring reusable Docker workflows across Windows, macOS, and Linux with schema synchronization, validation, and reliable cross-platform integration at scale.

Linguistix | *Validation, CI/CD*
- Developed and deployed a containerized Python REST API on AWS CodePipeline and ECS with automated CI/CD releases and validation checks, achieving 87% accuracy and production-grade reliability at scale.